

40. Suppose an $m \times n$ matrix A has n pivot columns. Explain why for each b in $\mathbb{R}^m$ the equation $Ax=b$ has at most one solution.

Solution: By defn, each of the n variables corresponding to the pivot columns are basic variables, i.e. there are no free variables.

- By Theorem 3, the matrix equation $Ax=b$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ b]$.
- By the Row Reduction algorithm and Theorem 1, the augmented matrix $[A \ b]$ can be converted to a unique rref.
- If in the rref form, there exists at least one row of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$, where $b \neq 0$, By Theorem 2 the system of equations is inconsistent, i.e. no solution set exists.
- If no such row exists, since there are no free variables, by Theorem 2, the system of eqns has a unique solution.

$\therefore$ For each b in $\mathbb{R}^m$ the equation $Ax=b$ has at most one solution $\square$